

2021-Ph-H1-Q3

Section: Our Dynamic Universe

Topic: Motion, Equations and Graphs

You're given a velocity–time graph.

From it, calculate:

- the acceleration
- the displacement

Solution:

- Acceleration:

$$a = \frac{10 - 15}{8} = -0.625 \text{ m/s}^2 \approx -0.63$$

- Displacement (area under v–t graph):

$$\text{Area} = \frac{1}{2}(15 + 10) \times 8 = 100 \text{ m}$$

Answer: A

Guidance for Students:

Find acceleration from gradient and displacement from area under the curve.

Revision Tip:

If the graph is a trapezium, use the trapezium area formula!

Check units and signs — especially for negative acceleration.